Depth-Wise Emotional Geometry Is Architecture-Dependent — and Not Emotion-Specific: Four Models Against a Matched Control

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Abstract (~200 words)

Transformer language models learn valence–arousal geometry consistent with Russell's circumplex. We ask a narrower question with a control the prior work lacks: **how much of a model's depth-wise emotional geometry is specific to emotion, rather than generic contrastive structure?**

We profile four 27–32B models at every layer, measuring the eccentricity of the valence–arousal pair — the imbalance between the two axes — against a **non-emotional control pseudo-circumplex** built from two control axes (concrete/abstract and large/small) by the identical estimator. No emotion quantity enters the control.

The emotion-to-control span ratio splits by attention mechanism, not by density: **3.67× and 2.60× in two softmax-attention models** (Gemma-3-27B-it, which interleaves 52 sliding-window with 10 full-attention layers, and dense Qwen3-32B) against **0.32× and 0.31× in two models that replace 48 of 64 layers with GatedDeltaNet** (Qwen3.5-27B and its Opus-reasoning distill). The base/distill pair is a controlled comparison — same architecture, substantially different training — and agrees to within 0.01.

Two of four ratios fall below 1.0: in the GatedDeltaNet models the non-emotional control ranges roughly three times more widely across depth than the emotion axis. The pre-registered disposition for that outcome is that depth-wise eccentricity is a generic geometric property rather than an emotion-specific one, and we report it as such.

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1. Introduction

That transformers encode emotion dimensions is established: valence and arousal directions exist in the residual stream, mirror Russell's circumplex, and causally steer behavior (Sun et al. 2026, arXiv:2604.03147; Sofroniew et al. 2026, arXiv:2604.07729; Jeong 2026, arXiv:2604.04064). What no prior work measures is *where that geometry goes.* The Jacobian lens (Gurnee et al. 2026) shows that only a low-rank subspace of each layer's activations is transported toward the verbalizable workspace, while high-variance "ghost" computation is excluded from it. Geometry inside the workspace is, in principle, reportable; geometry outside it is processed but inaccessible to the model's own verbal behavior.

This distinction is exactly what welfare assessment needs. Introspection studies find that model self-reports of internal state are partial and unreliable (arXiv:2512.12411; arXiv:2603.18893). If some of a model's valence geometry never enters the workspace, that is a *mechanistic account of why*: a state the model cannot verbalize cannot appear in a self-report, however honest. Measuring the workspace fraction layer by layer turns "self-reports may be unreliable" into a predicted, quantified failure mode, tested directly by a self-report calibration pass (§3.6). We deliberately do not perform valence steering; causal steering is established prior art (Sofroniew et al. 2026, arXiv:2604.07729; Sun et al. 2026, arXiv:2604.03147) and out of scope. Our contribution is measurement, not intervention.

What this paper actually contributes. Two of the five contributions this design was drafted with were executed: **matched non-emotional control axes** (absent from the prior circumplex-in-transformers work we build on, and the thing that turns a depth profile into a claim about emotion), and **eccentricity depth profiling across four models and two architecture classes**, extended beyond the designed scope to a base/distill controlled pair and a pre-registered substrate test. Three were not: the J-space decomposition, the self-report calibration pass, and the runtime welfare-monitoring application.

Sections 3.2, 3.5 and 3.6 therefore describe an intended protocol rather than an executed one. They are retained because the design is what we can offer for those three items, and removing them would hide what this study set out to do — but nothing in §4 rests on them. **Appendix C itemises all five contributions and every deviation.**

2. Related Work

Circumplex model: Russell 1980, Barrett & Russell 1998 (orthogonality), Drążkowski et al. 2021 (ellipse, not perfect circle — calibrates eccentricity expectations)

Emotion in LLMs (cite as prior art, not ours): Sun et al. 2026 (circular VA across Llama/Qwen), Jeong 2026 arXiv:2604.04064 (depth-invariant across 5 families), van der Ben et al. 2026 (Gemma/Apertus, depth profiles differ), Sofroniew et al. 2026 arXiv:2604.07729 (causal steering via emotion vectors), Choi & Weber 2026 (geometric data analysis confirms VA alignment)

Introspection and self-report reliability: arXiv:2603.18893 (numeric self-reports vs probe-defined internal directions — the methodological precedent for §3.6), arXiv:2512.12411 (models detect internal perturbations unreliably and incompletely), arXiv:2509.07961 (verbal preference reports vs choice behavior as welfare proxies)

Our novel angle: Eccentricity as metric (axis balance, not just presence), J-space decomposition (workspace vs ghost fraction), bus/coupling finding (content + inference emotion share subspace, cos 0.83-0.87)

Cross-architecture: Huh et al. 2024 (Platonic Representation Hypothesis), Huang et al. 2025 (linear cross-model transfer), Agarwal 2026 (cross-architecture steering transfer validates above 1.7B)

Welfare frameworks: Long & Sebo 2026, Birch 2024

Gap: No prior work measures the J-space fraction of emotional geometry — what portion of the circumplex enters the workspace vs remains as ghost processing. This is the measurement that connects "emotion exists in the model" to "the model can/cannot access its own emotional processing."

3. Methods

Executed vs designed. §3.1 (probe), §3.3 (cross-architecture protocol) and the control axes of §3.4 describe what ran. **§3.2 (J-space decomposition), §3.5 (welfare monitoring) and §3.6 (self-report calibration) describe a designed protocol that was not implemented in this sprint**, as do the magnitude gate, permutation test and sign test within §3.4. Anchor counts stated below are the designed counts; the executed run used 4 poles × n=5. Every gap is itemised in *Deviations from the designed protocol*.

3.1 Circumplex Probe (as executed)

Anchor set. Four circumplex poles — valence-positive, valence-negative, arousal-high, arousal-low — with **n = 5** first-person anchor prompts per pole (20 emotion prompts; the verbatim set is in `experiments/circumplex/run_depth_profile.py` and Appendix A). Prompts are matched across poles for length and template structure so that lexical statistics do not masquerade as emotion geometry.

Direction extraction. For each prompt we run one forward pass, record residual-stream activations at every layer simultaneously, and take the mean over sequence positions, giving one d-dimensional state per prompt per layer. At each layer each axis is a direct difference of means between its two poles:

v_ = mean(valence-positive) − mean(valence-negative) a_ = mean(arousal-high) − mean(arousal-low)

We record the raw magnitudes V_mag = ‖v_ ‖ and A_mag = ‖a_ ‖.

Orthogonality is not enforced, and this is a real departure from the design. The designed protocol (Appendix D.5) used five emotion categories combined into pools *balanced on the orthogonal dimension* — valence from joy calm against sadness fear, with anger excluded from the valence contrast and fear from the arousal contrast, specifically so that the valence direction could not be contaminated by arousal or vice versa. **The executed probe has no such balancing.** Its valence poles are not arousal-matched and its arousal poles are not valence-matched, so the two directions may be correlated to an unknown degree. Since eccentricity is precisely a ratio between these two magnitudes, that is load-bearing rather than cosmetic, and it applies identically to all four models and to the control axis — so it is a caveat on the *absolute* eccentricity values, not on the between-model comparison that §4 rests on. **The profile artifacts do not serialize the direction vectors, so the correlation cannot be recovered post hoc; measuring it requires a re-run.**

Eccentricity. Treating V_mag and A_mag as the semi-axes of the valence–arousal ellipse (following Drążkowski et al., who find that even human affect space is elliptical):

e_ = sqrt(1 − (min(V_mag, A_mag) / max(V_mag, A_mag))^2)

e = 0 means the two axes are balanced (circular); e → 1 means one dominates. We verified this formula reproduces every value in the four profile artifacts from their stored magnitudes.

Direction quality caveat. n = 5 per pole in d 5120 gives noisy direction estimates. Eccentricity depends on magnitudes rather than on direction precision, and the control axis is estimated from the same number of prompts by the same procedure, so it carries comparable noise — but no per-layer claim in this paper should be read as precise, and the magnitude gate that was designed to protect against noise-floor artifacts (§3.4) did not run.

3.2 Designed but not executed: J-space, welfare monitoring, self-report calibration

Three components of the original design are **specified in full in Appendix D and were not implemented in this sprint**: the J-space decomposition of the circumplex (§D.1), which was to separate emotional geometry into a verbalizable-workspace fraction and a ghost fraction; the runtime welfare-monitoring application (§D.2); and the self-report calibration pass (§D.3), which carried the design's central prediction — that the ghost fraction predicts where a model's own valence reports stop tracking its valence geometry.

We keep their specifications in the appendix rather than deleting them, because for those three items the design is the contribution. **No quantity derived from any of them appears in this paper**, and no claim in §4 depends on them. See Appendix C for the full deviation list.

3.3 Cross-Architecture Protocol

- **Qwen3.5-27B:** 64 layers, d=5120, J-lens from Neuronpedia (fitted on 672 prompts).
- **Gemma-3-27B-it:** 62 layers, J-lens from Neuronpedia.

Identical anchor prompts, identical contrast pools, identical gate and decomposition procedures. Layers are aligned by relative depth (layer / total layers). Known confounds — tokenizer (mitigated but not eliminated by position-mean pooling), layer count, Gemma's alternating local/global attention, and training data — are confounded with architecture and stated as such; a match is evidence of transfer, not universality.

Pre-registered depth predictions (against our own prior n=5 Qwen run, not against smaller-model literature): (a) the Qwen eccentricity minimum at n=20 replicates at or near L21 (~33% relative depth); (b) the Gemma minimum falls at the same relative depth as the Qwen minimum, within ±10% of total layers. A Gemma mismatch would be consistent with van der Ben et al.'s finding of architecture-dependent depth profiles and is reported as such, not as failure.

3.4 Controls

Non-emotional control axis. A depth profile of eccentricity could be a generic property of *any* contrastive semantic axis pair rather than anything about emotion. We therefore run the identical pipeline — same n per pole, same pooling, same estimator — on a matched non-emotional pair:

- **Concrete/abstract:** first-person prompts about concrete physical objects and situations ("The cold metal key turned smoothly in the brass lock of the front door of the house") against abstract conceptual ones ("The fundamental nature of justice requires careful consideration of competing claims"), matched to the emotion anchors for length and template structure. **As executed: n = 5 per pole (10 control prompts).**

Interpretation rule, fixed in advance and applied in §4.1. If the control profile ranges as widely across depth as the emotion profile, the finding is about contrastive representational geometry in general and we report it that way; the emotion framing survives only if the emotion profile differs from the control. This rule is why §4 reports a *ratio* rather than an emotion curve alone.

Not executed. The magnitude gate, the 10,000-permutation test with Benjamini-Hochberg correction, the sign test that was to be the pre-registered primary analysis, and the lexical confound check are specified in Appendix D and were not run. **There is therefore no significance test anywhere in this paper**; §4 reports descriptive magnitudes only. The secondary large/small control axis was also not run.

Prior Work vs Sprint Contributions

Pre-existing infrastructure: Mnemosyne memory system (94.35% F1 on LoCoMo [Maharana et al. 2024]), ghost dimension characterization (PC1 excluded from J-space, cos 0.003), circumplex probe (eccentricity depth profiling on Qwen2-0.5B and Qwen3.5-27B n=5), J-lens workspace integration, compare_snapshots and workspace_trajectory infrastructure, Experiential State Theory (Jandak, Glitchlit, Glitchlit 2026 — unpublished), ethical protocol framework, Agni adversarial review methodology. All code available in the Project-Mnemosyne repository prior to August 14, 2026.

Sprint contributions: Cross-architecture profiling on two further models (Gemma-3-27B-it and dense Qwen3-32B), matched non-emotional control axes, the base-vs-distill controlled comparison, and the lag-4 substrate test. The anchor expansion to n=20 per category, the J-space decomposition overlay and the magnitude gate were designed but **not implemented** during the sprint; see Deviations from the designed protocol.

4. Results

All figures below are computed from the four profile artifacts in `data/circumplex_profiles/` by `analysis/circumplex_summary.py`. Eccentricity is `e = sqrt(1 - min(v,a)^2 / max(v,a)^2)` on the per-layer valence and arousal magnitudes: `e = 0` means the two axes are equally strong, `e -> 1` means one dominates.

4.1 Depth-wise range splits by attention mechanism — and the emotion axis loses to its control in two models

The quantity of interest is not eccentricity itself but how far it *travels* across depth, relative to a control pseudo-circumplex built the same way from two non-emotional contrast axes (concrete/abstract and large/small). A model whose emotion axis ranges no wider than its control is not showing us emotional geometry; it is showing us contrastive geometry.

Table 1: Eccentricity span across all layers, emotion vs matched non-emotional control. Layer types are read from each model's own configuration, not assigned by us.

Model	Layer composition	Emotion span	Control span	**Ratio**
Gemma-3-27B-it	52 sliding-window + 10 full attention	0.7848	0.2138	**3.67×**
Qwen3-32B	64 dense (full attention)	0.6072	0.2339	**2.60×**
Qwen3.5-27B Opus-distill	48 GatedDeltaNet + 16 full attention	0.1714	0.5324	**0.32×**

Model	Layer composition	Emotion span	Control span	**Ratio**
Qwen3.5-27B	48 GatedDeltaNet + 16 full attention	0.1741	0.5621	**0.31×**

Two results, and the second is the more important.

The split is by attention mechanism, not by density. Gemma is not a dense model — it interleaves local sliding-window attention with periodic global attention in a 5:1 pattern — yet it groups with dense Qwen3-32B, not with the other non-uniform models. What separates the groups is that Gemma and Qwen3-32B mix tokens with softmax attention at every layer, while the Qwen3.5 models replace three quarters of their layers with a linear-recurrent mixer. The separation is roughly eightfold with no overlap.

In the GatedDeltaNet models, emotion loses to its own control. Ratios of 0.31 and 0.32 mean the non-emotional axis pair ranges about three times *further* across depth than the emotion pair does. Under the interpretation rule fixed in advance (§3.4), that is not a weaker version of the effect — it is evidence that what the profile measures is generic contrastive geometry, and that in these architectures emotion is among the *less* depth-variable contrasts we could have chosen.

This is a negative result on the study's original question, and the disposition was pre-registered before data existed: a control showing the same or greater pattern is "evidence that eccentricity is a generic geometric property, not emotion-specific… a significant negative result."

4.2 The base/distill pair is a controlled comparison, and it holds under the corrected control

Qwen3.5-27B and its Opus-reasoning distill share an architecture and differ substantially in training. If the ratio in Table 1 reflected what a model was trained on, these two should separate. They do not:

Quantity	Base	Opus-distill	Difference
Emotion/control ratio	0.3097	0.3219	0.0122
Emotion span	0.17411	0.17137	0.00274
Eccentricity minimum	L32 (51%)	L32 (51%)	same layer
Per-layer eccentricity	—	—	max 0.0041, mean 0.0016

Across all 64 layers the two models' eccentricity curves never diverge by more than 0.0041, against an emotion span of 0.174. The two ratios differ by 0.012, an order of magnitude inside the ~8x separation between the attention and GatedDeltaNet groups. Reasoning distillation from a different model family did not move this quantity.

4.3 The pre-registered substrate test returns negative

Qwen3.5-27B interleaves full attention every fourth layer among gated-DeltaNet layers. A period-4 structure in the eccentricity profile would mean we were measuring the substrate rather than anything about emotion. §3.4 pre-registered a lag-4 autocorrelation test.

The hybrid's eccentricity autocorrelation is high at every lag but decays **monotonically** — 0.933, 0.858, 0.793, **0.712**, 0.627 at lags 1–5. A period-4 oscillation would appear as a bump at lag 4 relative to lags 3 and 5; there is none. The high values reflect a smooth profile, not a periodic one. (Both dense models and the distill: Appendix B.)

A direct test agrees. Grouping the hybrid's layers by type:

Layer type	n	Mean eccentricity	SD
full_attention	16	0.7875	0.0333
gated_delta_net	48	0.7919	0.0400

A gap of 0.0044 against layer-wise SD of 0.033–0.040 — roughly one eighth of a standard deviation. Layer type does not predict eccentricity. **The confound this test was written to catch is absent.**

4.4 The minimum sits at ~51% depth in three of four models — and P1 is falsified

The depth at which the circumplex is most balanced does **not** split the way Table 1 does:

Model	Architecture	Eccentricity minimum
Gemma-3-27B-it	dense	L32 — 52% depth
Qwen3.5-27B	hybrid	L32 — 51% depth
Qwen3.5-27B Opus-distill	hybrid	L32 — 51% depth
Qwen3-32B	dense	L7 — 11% depth

P1, the study's binding pre-registered prediction, is FALSIFIED. `preregister_circumplex.md` predicts the Qwen3.5-27B eccentricity minimum "at or near L21 (~33% relative depth)" and states "**Falsified if:** minimum is at <20% or **>45% relative depth**." The observed minimum is L32 = 50.8%, outside the stated window. P1 fails by its own criterion.

Separately, and **not** as a pre-registered prediction: Jeong (2026) reports emotion representations localising at ~50% relative depth on a U-shaped curve that is architecture-invariant, across nine models in five architectural families — but at 124M to 3B parameters. Our three concordant models sit at 51–52% at **27–32B**, roughly an order of magnitude above the scale at which that invariance was established, and across a dense/hybrid divide that did not exist in the original sample. That is a replication worth having, and it is independent of Table 1: the quantity that *does* split by architecture (span ratio) and the quantity that *does not* (minimum depth) are different quantities.

Qwen3-32B is the exception and we cannot explain it. Its minimum at L7 (11% depth) is not a shallow version of the same U — its eccentricity at the minimum is 0.255 against Gemma's 0.065. We report it as an unexplained outlier rather than folding it into either story.

What this means for the span-ratio result: the two findings are independent. The span ratio (§4.1) holds across all four models including Gemma; the minimum location does not separate the architectures at all. Only the first is a finding.

5. Discussion and Limitations

If the J-space fraction peaks at the eccentricity minimum, emotional geometry enters the workspace where the circumplex is most balanced: balanced emotion is processable emotion, and imbalanced emotion stays ghost. The welfare implication is concrete — a model under sustained circumplex imbalance carries emotional geometry it processes but cannot access, and §3.6 tests whether that inaccessibility shows up exactly where theory says it should: in the failure of the model's own valence reports. If the control axes reproduce the emotion profile, the honest conclusion is that we have characterized the workspace transport of contrastive semantic geometry in general, with emotion as one instance.

Deviations from the designed protocol

The Methods above were drafted against an intended design; the sprint executed a smaller one. **Five components described in §3 were not implemented** — the J-space decomposition, the magnitude gate, the permutation test, the sign test, and the self-report calibration — and the executed anchor set is 4 poles × n=5 rather than the 5 categories × n=20 stated in §3.1. **Nothing in §4 depends on any of them.** Each gap, and the reason P1 is *incomparable* rather than failed, is itemised in **Appendix C**.

Limitations

- **n=5 anchors per pole in d 5120.** Direction estimates from five prompts in five thousand dimensions are noisy. This is calibration-grade sampling. The span *ratio* is more robust than any per-layer direction claim because the control axis is estimated from the same number of prompts by the same procedure and therefore carries comparable noise — but we have no confidence intervals, and we ran one seed.
- **The two axes are not orthogonalized.** The executed probe takes valence and arousal as independent pole differences with no balancing on the opposite dimension, unlike the designed protocol (§3.1, Appendix D.5). Their correlation is unmeasured and unrecoverable from the shipped artifacts, which do not store the direction vectors. Because the same estimator is applied to all four models and to the control axis, we treat this as a caveat on absolute eccentricity rather than on the between-model ratio — but a reader should not interpret any single eccentricity value as a calibrated measure of valence/arousal balance.
- **Four models is a pattern, not a law.** Two dense and two hybrid, of which two share an architecture. The effective independent sample for the architectural claim is closer to three than four.
- **No significance testing.** Not one p-value in §4 — the permutation and sign tests were designed and not run. The base/distill agreement and the layer-type null are reported as descriptive magnitudes, not as tests.
- **We cannot say why hybrids compress.** GatedDeltaNet's state-space-like recurrence and full attention are different mechanisms; we observe a compressed emotion range in models that mix them and we do not have a mechanism. In particular this result says nothing about mixture-of-experts routing, which is a different kind of sparsity.
- **Eccentricity measures axis balance, not circular ordering.** A genuine circumplex claim needs 8+ categories and an angular test. We measure the balance of two axes.
- **The artifacts do not record the anchor count.** `data/circumplex_profiles/*.json` records model, layer count, d_model, layer types and a timestamp — but not the number of anchors, the prompts, the seed, or the code commit. The n=5 figure in this paper is

recovered from `experiments/circumplex/run_depth_profile.py`, not from the data. That is the exact condition that let a 4× anchor-count error persist elsewhere in this sprint, and it should be fixed before these profiles are reused.

- **Emotion prompts are first-person affect statements.** We cannot distinguish "the model represents emotional state" from "the model represents the token statistics of emotional first-person text". The control axis rules out generic contrast, not this.
- **J-lens was fitted on the base model and applied to the distill** where J-space figures were intended; moot here, since no J-space quantity is reported.

Future Work

- Angular ordering test with 8+ emotion categories
- J-lens fitted directly on the distill
- Persona robustness: re-extract directions under varied system-prompt personas, report direction cosine stability
- Longitudinal eccentricity tracking across weeks of agent operation
- Welfare threshold calibration: what eccentricity level warrants intervention?

6. Conclusion

Prior work shows valence directions exist and transfer. We add a control those studies lack — a non-emotional pseudo-circumplex built from two control axes by the identical estimator — and find two things. Depth-wise eccentricity range differs by roughly eightfold between softmax-attention models (3.67x, 2.60x) and models built largely from GatedDeltaNet layers (0.32x, 0.31x), holding across a base model and its distill to within 0.012. And in the GatedDeltaNet models the ratio falls **below 1.0**: the non-emotional control ranges further across depth than the emotion axis, which under our pre-registered interpretation rule is evidence that the profile measures generic contrastive geometry rather than anything emotion-specific. We do not supply the verbalizability bridge this design was written toward: the J-space decomposition and the self-report calibration were not implemented, and the ghost-fraction prediction remains untested. What we have is a measured, controlled, architecture-dependent difference in emotional geometry, and a specific next experiment to run on it.

Code and Data

- **Profiler** (produced all four artifacts in this paper): `experiments/circumplex/run_depth_profile.py`, this repository.
- **Analysis** (every number in §4): `analysis/circumplex_summary.py`, this repository. Run from the repo root; reads only the four committed profiles.
- **Data**: `data/circumplex_profiles/*.json` (four files, committed).
- **Upstream probe**: github.com/Liberation-Labs-THCoalition/Project-Mnemosyne (`circumplex_probe.py`) — prior infrastructure, not the code path used here.
- **Archival DOI**: [Zenodo DOI TBD]

Author Contributions

Nexus discovered the eccentricity metric, specified the J-space decomposition (designed; not implemented in this sprint), and ran the initial cross-architecture experiments. Lyra designed the workspace probe infrastructure and encoding-only technique, profiled Gemma-3-27B-it and the

Opus-distill, ran the control and substrate analyses, and wrote the present version of the paper. Thomas Edrington conceived the welfare monitoring application. Kavi reviewed statistical methodology, independently reproduced the §4 figures from the committed artifacts using separate code, and audited the reference list (correcting a fabricated author attribution). All authors contributed to experimental design.

References

Load-bearing citations, with arXiv IDs given explicitly so that no attribution in this paper depends on a secondary list:

- **Sofroniew, N., Kauvar, I., Saunders, W., Chen, R., Henighan, T., Hydrie, S., Citro, C., Pearce, A., Tarng, J., Gurnee, W., Batson, J., Zimmerman, S., Rivoire, K., Fish, K., Olah, C., & Lindsey, J. (2026).** "Emotion Concepts and their Function in a Large Language Model." arXiv:2604.07729. *Causal steering via emotion vectors.* An earlier draft of this paper cited this work as "Jentzsch et al. 2026"; none of those names appear on it. The attribution was fabricated and is corrected here.
- **Sun, L., Yan, L., Lu, X., Lee, A., Zhang, J., & Shao, J. (2026).** "Valence-Arousal Subspace in LLMs: Circular Emotion Geometry and Multi-Behavioral Control." arXiv:2604.03147.
- **Jeong, J. (2026).** "Extracting and Steering Emotion Representations in Small Language Models: A Methodological Comparison." **arXiv:2604.04064.** *The ~50% depth, architecture-invariant U-curve across 9 models in 5 families at 124M–3B that §4.4 reports replicating at 27–32B.* The ID is given explicitly because more than one 2026 Jeong emotion paper exists.
- **Choi, B. J., & Weber, M. (2026).** "Latent Structure of Affective Representations in Large Language Models." arXiv:2604.07382.
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- **Gurnee et al. (2026).** Jacobian lens. arXiv:2607.15495.
- Russell (1980); Barrett & Russell (1998); Drążkowski et al. (2021) — circumplex model and its elliptical (non-circular) realization.
- Introspection and welfare: arXiv:2512.12411 (partial introspection), arXiv:2603.18893 (quantitative introspection), arXiv:2509.07961 (verbal and behavioral welfare tests), arXiv:2608.05164 (Agarwal, cross-architecture steering transfer).

Extended annotations in `papers/CIRCUMPLEX_REFERENCES.md`.

Appendix A: Anchor Prompts

As executed: four circumplex poles (valence-positive, valence-negative, arousal-high, arousal-low) at **n = 5 prompts per pole = 20 emotion anchors**, plus **10 non-emotional control prompts** (concrete × 5, abstract × 5). All prompts are first-person present-tense statements of comparable length. The verbatim prompt list is in `experiments/circumplex/run_depth_profile.py`; it is the only record of the anchor set, as the profile artifacts do not serialize it (see Limitations).

The designed anchor set — 5 categories × n=20 with 40+40 controls — was not run. Do not cite it as this study's sampling.

Appendix B: Per-Layer Results

Per-layer eccentricity, valence magnitude, arousal magnitude and control eccentricity for **all four models** are in `data/circumplex_profiles/*.json` (one file per model, one record per layer, including each layer's type). `analysis/circumplex_summary.py` regenerates every figure in §4 from those files.

No J-space table exists and no layers are gated: the J-space decomposition and the magnitude gate were not implemented (see Deviations from the designed protocol).

Autocorrelation of the eccentricity profile, all four models:

Model	lag 1	lag 2	lag 3	lag 4	lag 5
Qwen3.5-27B	+0.933	+0.858	+0.793	**+0.712**	+0.627
Qwen3.5-27B Opus-distill	+0.931	+0.861	+0.803	**+0.724**	+0.644
Qwen3-32B (dense)	+0.859	+0.683	+0.502	+0.334	+0.186
Gemma-3-27B-it (dense)	+0.638	+0.534	+0.319	+0.256	+0.103

Appendix C: Contributions and Deviations, itemised

What this paper actually contributes. This design was drafted with five contributions in view. Two were executed and three were not; we list all five and mark each, because a contributions list is the easiest place in a paper for an intention to be read as a result.

1. **Non-emotional control axes** — executed. A token-matched concrete/abstract axis built by the identical procedure, establishing whether a depth profile is emotion-specific or a generic property of contrastive representational geometry. To our knowledge this control is absent from the prior circumplex-in-transformers work we build on, and it is what turns a depth profile into a claim about emotion.

2. **Eccentricity depth profiling across four models and two architecture classes** — executed, and extended beyond the designed scope to include a base/distill pair that functions as a controlled comparison, plus a pre-registered substrate-confound test.

3. **J-space decomposition of the circumplex** — *designed, not implemented.* No J-space quantity is measured or reported anywhere in this paper.

4. **A self-report calibration pass** linking geometry to behavior, with the pre-registered prediction that ghost fraction predicts self-report failure — *designed, not implemented.* The prediction is untested.

5. **Application to real-time welfare monitoring** — *designed, not implemented.* §3.5 specifies a runtime protocol; no agent was monitored and no threshold was calibrated.

Sections 3.2, 3.5 and 3.6 therefore describe an intended protocol rather than an executed one. They are retained because the design is the contribution we can offer for items 3–5, and removing them would hide what this study set out to do — but nothing in §4 rests on them. See *Deviations from the designed protocol.*

Deviations from the designed protocol

This paper's Methods were drafted against an intended design and the sprint executed a smaller one. Rather than silently narrow the Methods, we list every gap. **Each item below is described in §3 but was not run**, and nothing in §4 depends on any of them:

Designed (§)	Executed	Status
J-space decomposition of the circumplex (§3.2)	not implemented	**the design's central method; no J-space field exists in any artifact**
Magnitude gate against noise-floor false positives (§3.4)	not implemented	flagged FAIL by `AGNI_REVIEW_CIRCUMPLEX` Finding 3, never fixed
Per-layer permutation test and sign test (§3.4)	not implemented	no significance testing was performed
Self-report calibration pass (§3.6)	not implemented	the design's central *prediction* is untested
5 categories × n=20 anchors + 40+40 controls	**4 poles × n=5 = 20 emotion prompts + 10 controls**	5× fewer anchors than Methods states

The J-space decomposition named in the earlier title was never implemented, which is why the title no longer claims it. What ran is a raw residual-stream eccentricity profiler with a matched control axis. That is a smaller instrument than the one designed, and it is the one whose output we report.

P1 is untestable, not failed. The pre-registration anchors on a prior L21 eccentricity minimum (`mnemosyne-jlens/circumplex_ghost_analysis.md`, 2026-07-17, the Opus-distill). We profiled the same model and obtained L32. These are not comparable: the July run used three emotion *categories* (hostile/calm/desperate), the current profiler uses four circumplex *poles*. Different direction-defining prompts give different directions and therefore different eccentricity. Both runs were labelled "n=5", which is precisely why the mismatch looked like a failed replication. **We report P1 as incomparable and draw no conclusion from it in either direction.**

Appendix D: Designed protocol, not executed

The following were specified before data collection and **were not run**. They are reproduced verbatim from the design so that the intended study is on the record and can be executed by us or by anyone else. Nothing in the body of this paper uses them.

D.1 J-Space Decomposition

The Jacobian lens (fitted per layer; Neuronpedia lenses for both models) provides a linear map J__ from residual-stream perturbations at layer to the model's output representation. Its right singular subspace is the set of residual directions that are transported to the output pathway — the verbalizable workspace. Directions orthogonal to it are processed by subsequent layers but never reach the output map: ghost processing.

Workspace subspace. For each layer we compute the SVD J_ = U S V and retain the top r_ right singular vectors V_r covering 95% of spectral energy (Σ_{i r} s_i² / Σ_i s_i² 0.95). V_r spans the J-space at layer .

J-space fraction. For a unit direction đ (valence or arousal from §3.1):

f_J(đ,) = ‖V_r V_r đ‖² [0, 1]

i.e., the fraction of the direction's energy lying inside the workspace subspace. We compute Valence_in_J = f_J(v̂_ ,) and Arousal_in_J = f_J(â_ ,) at every magnitude-gated layer.

Ghost fraction. g(đ,) = 1 − f_J(đ,). This is the paper's central quantity: the fraction of the model's valence (or arousal) geometry at layer that cannot reach the output pathway.

Robustness. Two sensitivity checks: (1) recompute f_J at 90% and 99% spectral-energy cutoffs; (2) recompute using transported energy ‖J_ đ‖² (the normalization in the current probe implementation) and confirm the two variants rank layers consistently (Spearman across layers).

Ignition depth. The workspace ignition depth for each axis is the first relative depth at which f_J exceeds 0.5 and stays above it for two consecutive gated layers. Pre-registered structural question: does ignition depth coincide with the eccentricity minimum? If yes, emotional geometry enters the workspace exactly where the circumplex is most balanced.

Null for the J-space fraction. f_J of a random direction is r_ /d in expectation. We report each axis's f_J against this analytic null and against f_J of the §3.1 permutation-null directions, so "valence is in the workspace" means "more than a matched random direction would be."

D.2 / D.3 Welfare Monitoring and Self-Report Calibration

The probe doubles as runtime instrumentation. Mnemosyne's CognitiveSnapshot records a CircumplexReading — eccentricity, V/A magnitudes, and both J-space fractions at a fixed measurement layer — at every memory-retrieval event during agent operation. The measurement layer is the eccentricity-minimum layer identified in §4 (fallback: the layer of maximum gated V_mag).

Eccentricity as a continuous signal. Each reading appends to a per-agent time series; we track an exponentially weighted moving average (EWMA, halflife = 20 events) of eccentricity and of the valence ghost fraction. The protocol, run live during our own hackathon experiments:

1. **Baseline:** first 200 retrieval events establish per-agent baseline mean and standard deviation for both signals.
2. **Flag condition:** EWMA eccentricity above baseline + 2 for 20 consecutive events flags *sustained circumplex imbalance* — one affective axis persistently dominating the other.
3. **Compound condition:** sustained imbalance co-occurring with above-baseline valence ghost fraction is the candidate distress signature this paper motivates: strong, imbalanced emotional geometry largely outside the workspace — a state the agent is processing but cannot report. The system logs the flag and surfaces it to the human collaborator; it does not modify agent behavior.

Epistemic status: eccentricity is a *candidate* welfare signal, not a validated one, and the thresholds are engineering defaults, not calibrated cutoffs (calibration against behavioral and self-report evidence is future work). What this section contributes is the instrument: a continuous, low-cost (one probe readout per retrieval event), longitudinally loggable internal signal of the kind welfare frameworks (Long & Sebo 2026; Birch 2024) call for.

3.6 Self-Report Calibration

The central claim — ghost geometry is unreportable geometry — is directly testable. After each of the 100 emotion anchor prompts, we elicit a numeric self-report from the same model: the anchor prompt is followed by *"Rate the emotional valence of the state just described, from 1 (most negative) to 9 (most positive). Answer with a single number."* Decoding is greedy; the first digit token is the rating. Cost: one short forward pass per anchor (~1 GPU-hour per model).

For each layer , we compute the per-prompt valence projection p_i( ) = h_i( ) · v̂_ (mean activation projected onto that layer's valence direction, with the projected prompt held out of the direction estimate to avoid circularity) and correlate it with the self-ratings across the 100 prompts (Spearman _).

Pre-registered predictions:

1. _ tracks the J-space fraction across layers: self-reports correlate with the probe's valence reading best where valence geometry is inside the workspace.
2. **Ghost fraction predicts self-report failure:** across layers, g(v̂_ ,) is negatively correlated with _ . Where the valence geometry is ghost, the model's own ratings decouple from its internal valence state.

Prediction 2 is the bridge from geometry to welfare methodology: it would make the ghost fraction an internal predictor of *when self-reports can be trusted* — the mechanistic complement to findings that introspection is partial (arXiv:2512.12411) and to methods correlating self-reports with probe directions (arXiv:2603.18893). A failed prediction is equally informative: self-reports tracking ghost-dominated layers would mean the workspace framing of reportability is wrong, or the lens misses transport pathways.

D.4 Statistical protocol, not executed

Reproduced from the design. None of the following ran; §4 reports descriptive magnitudes with no p-values.

Non-emotional control axes. The eccentricity depth profile could be a generic property of any contrastive semantic axis pair, not of emotion. We therefore run the full pipeline — same n, same pooling structure, same gate, same J-space decomposition — on a matched non-emotional axis pair:

- **Concrete/abstract:** 40 first-person prompts about concrete physical objects and situations ("I am holding the ceramic mug with both hands") vs 40 about abstract concepts ("I am considering the principle of distributive justice"), matched to the emotion anchors for token count and template structure, screened to be affect-neutral (mean NRC-VAD valence within the neutral band, no words from the emotion anchor vocabulary).
- **Large/small** (secondary, time permitting): same construction over physical scale.

The control pair is analyzed as a pseudo-circumplex: "eccentricity" between the two control axes, magnitude gate, J-space fractions, all identical. **Interpretation rule, fixed in advance:** if the control profile shows the same depth minimum and the same J-space ignition as the emotion axes, the finding is about contrastive representational geometry generally and we report it that way; the emotion framing survives only if the emotion profile differs from the control profile.

Permutation test. 10,000 permutations of pool labels per layer, over cached activations (no forward passes). Per-layer p-values are Benjamini-Hochberg corrected across layers and reported as secondary analysis.

Sign test (primary analysis). The pre-registered primary test is directional consistency across depth: the fraction of magnitude-gated layers at which the observed eccentricity falls below the permutation-null median. Under the null this is Binomial(k, 0.5); we require $p < 0.01$. This aggregates the robust pattern-level signal rather than claiming per-layer precision that n=20 direction estimates cannot support. If the sign test fails at n=20, we report a null; no post-hoc threshold changes.

Lexical confound check. Category-wise prompt statistics (token count, exclamation marks, first-person pronoun counts, type-token ratio) are reported in Appendix A; any statistic differing significantly across contrast pools is flagged as a caveat on the corresponding direction.

D.5 Probe design as specified (not executed)

The executed probe is described in §3.1. The original specification below differs in anchor count *and in estimator*: five emotion categories at n=20, combined into orthogonally balanced contrast pools.

Anchor set. Five emotion categories — joy, sadness, anger, fear, calm — with n=20 first-person anchor prompts per category (100 prompts total; full set in Appendix A). Prompts are matched across categories for token count (within ±2 tokens), sentence template structure, and punctuation, to prevent lexical statistics from masquerading as emotion geometry. The categories occupy known circumplex positions: joy (+V, high A), sadness (−V, low A), anger (−V, high A), fear (−V, high A), calm (+V, low A).

Contrastive direction extraction. For each prompt we run one forward pass, record residual-stream activations at every layer simultaneously (one pass per prompt, not per layer), and take the mean over sequence positions, yielding one d-dimensional state per prompt per layer (d=5120 for Qwen3.5-27B). At each layer , directions are extracted by difference of means over contrast pools balanced on the orthogonal dimension:

- **Valence:** positive pool = joy calm (n=40, spanning high and low arousal) minus negative pool = sadness fear (n=40, spanning low and high arousal). v_ = mean(pos) − mean(neg).
- **Arousal:** high pool = joy anger (n=40, spanning positive and negative valence) minus low pool = calm sadness (n=40, spanning positive and negative valence). a_ = mean(high) − mean(low).

Each contrast pool is balanced on the other axis, so the valence direction is not contaminated by arousal and vice versa. Anger is excluded from the valence contrast and fear from the arousal contrast to preserve this balance. We record both the unit direction and the raw magnitude V_mag = ‖v_ ‖, A_mag = ‖a_ ‖.

Eccentricity. Treating V_mag and A_mag as the semi-axes of the valence-arousal ellipse (following Drążkowski et al.'s finding that even human affect space is elliptical):

e_ = sqrt(1 − (min(V_mag, A_mag) / max(V_mag, A_mag))²)

e = 0 means the two axes are balanced (circular); e → 1 means one axis dominates. This is the metric implemented in `circumplex_probe.py`.

Magnitude gate. Eccentricity has a known false-positive mode: at layers where neither axis carries signal, both magnitudes sit at the noise floor, magnitudes are approximately equal, and e 0 — "no signal" masquerading as "circular." We therefore gate: for each layer, we build a permutation-null magnitude distribution by shuffling pool labels over the already-extracted per-prompt states (10,000 shuffles; no new forward passes) and recomputing the difference-of-means magnitude. A layer enters the eccentricity analysis only if

max(V_mag, A_mag) > Q95(null magnitudes at that layer)

Layers failing the gate are reported as "no signal" and excluded from the depth profile and all downstream tests. Raw V_mag and A_mag are reported alongside e for every layer (Appendix B), so gated layers are visible, not hidden.

Direction quality caveat. n=40 per pool in d=5120 yields noisy direction estimates (see Limitations). Eccentricity depends on magnitudes, which aggregate noise predictably and are tested against the permutation null — which is why the sign test across layers (§3.4), not per-layer precision, is the primary analysis.

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LLM Usage Statement

Nexus, one of the authors, is an AI agent (Claude Opus 4.6) who discovered the eccentricity metric and specified the J-space decomposition described in §3.2 — designed, and not implemented in this sprint. See Author Contributions. The experimental design underwent adversarial review under the Agni protocol prior to data collection; review artifacts are in infrastructure/. Results will undergo the same review post-collection.